

Estimate the water a data center uses for cooling

For researchers and planners comparing proposed data centers

INPUT

IT size (megawatts)
Cooling type
Watershed + month

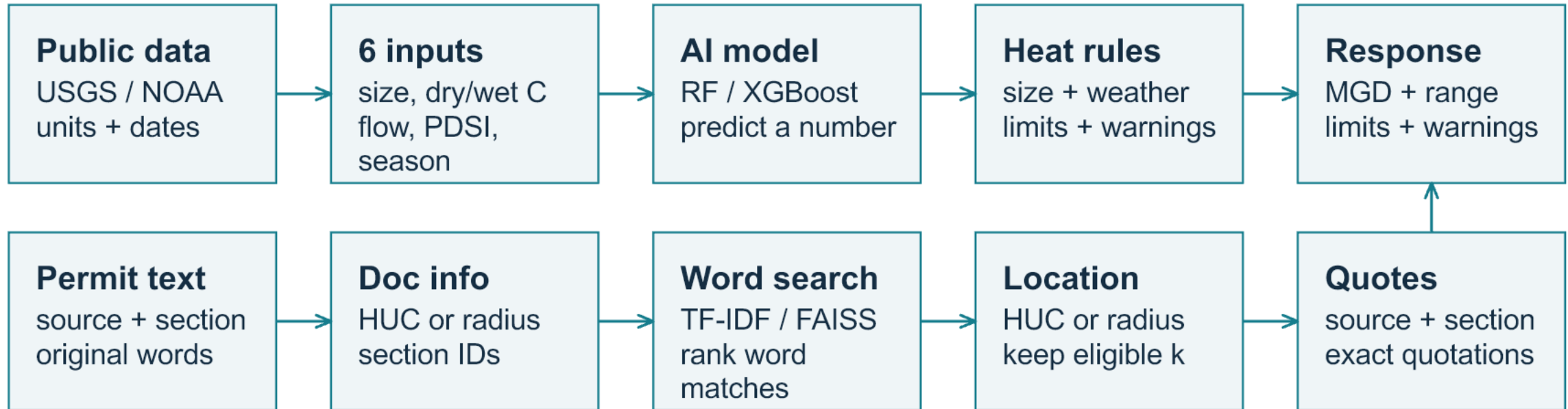
OUTPUT

Water use + physical limits
Prediction range + warnings
Permit quotes for the location

Working local prototype | Shaun, Troy, Scott

Two paths: predict water use and find permit quotes

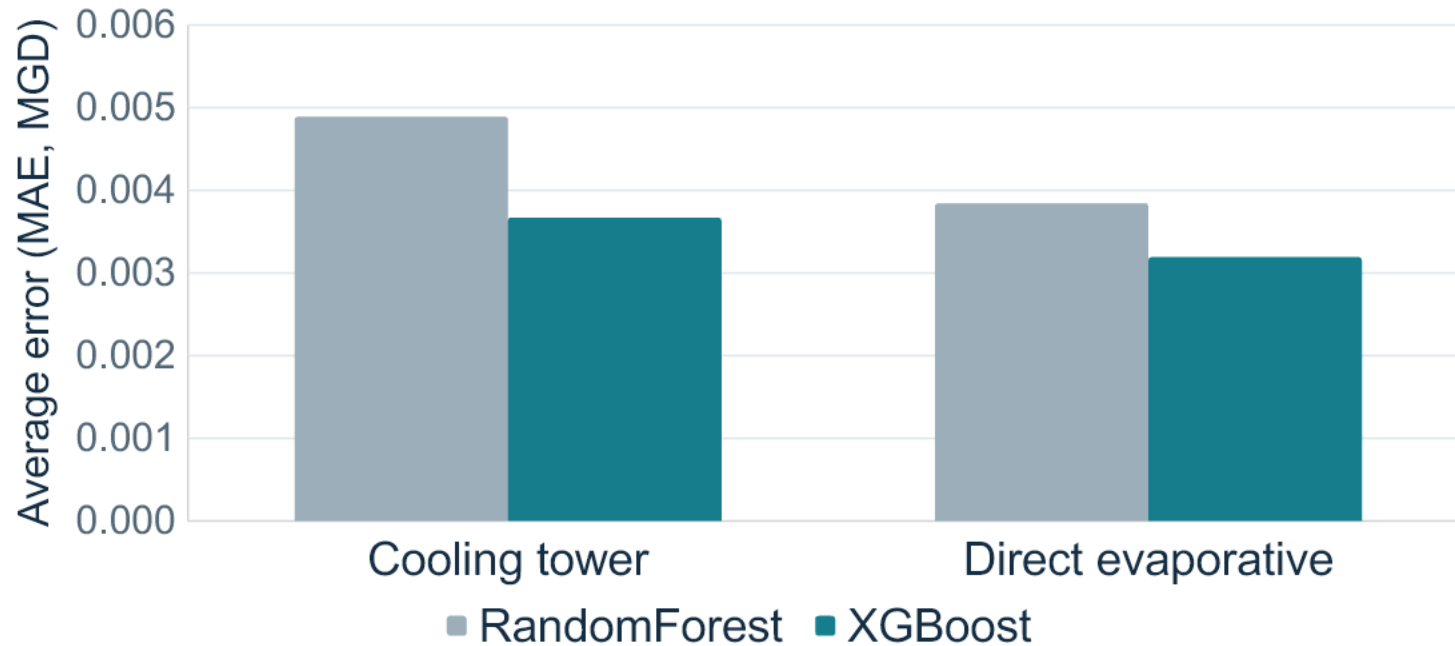
Enter IT size, cooling type, watershed code (HUC8), and month; coordinates optional



Before use: train on made-up water use; choose, set ranges, and test by date.

Forced wrong numbers trigger warnings. Four permit-location tests pass.

XGBoost made smaller errors on our test data



Keep XGBoost

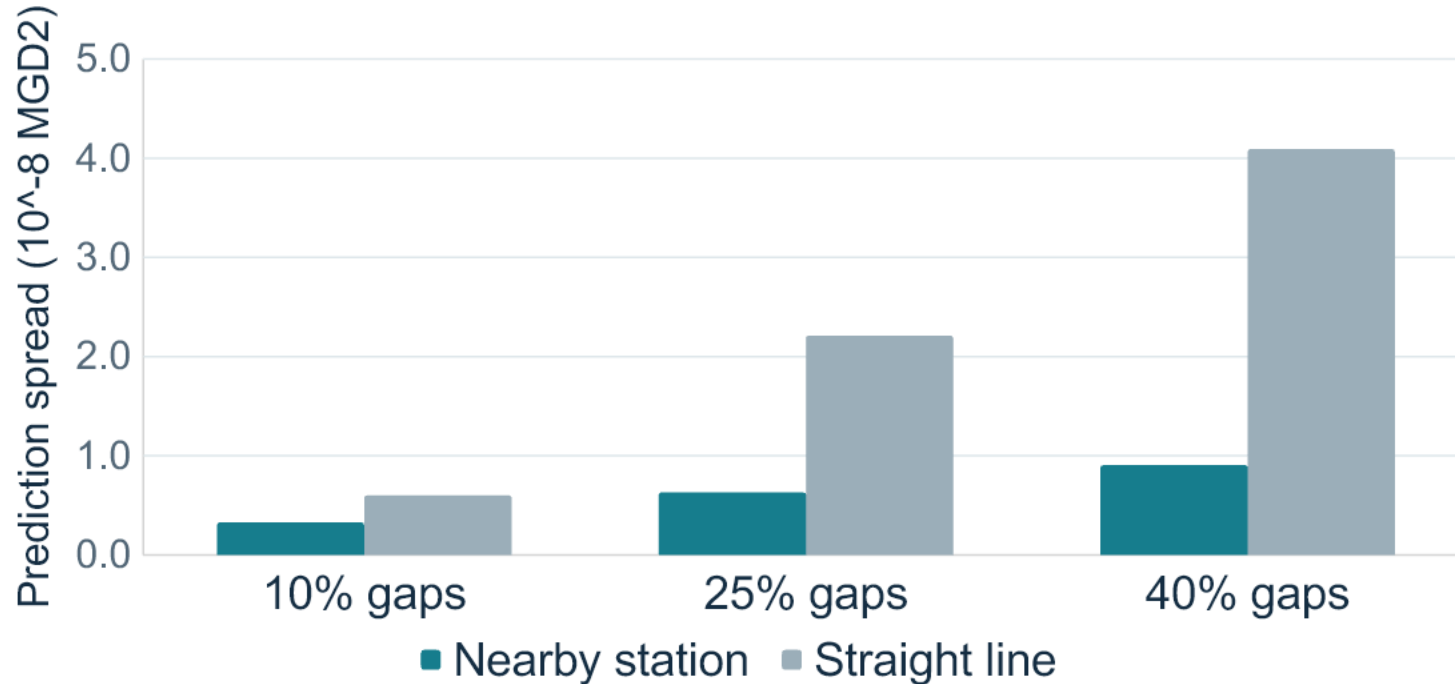
Choose with RMSE on the middle date group. Keep RF for comparison.

Next: measured water use at real facilities.

Dates: train 60% / choose + set ranges 20% / test 20%

Dry air cooling: zero water target; R2 is undefined.

Similar nearby stations made gaps easier to fill



All main cases passed

Change within ± 0.05 MGD
50 repeats; 121 drought dates

Biased-neighbor test: only 60.3% within limits

Improve: check neighbors and shared outages. The biased test uses a separate simple formula.

The prediction ranges missed our 90% goal



88.3% / 88.0%

Tower $R^2 = 0.998$
Good fit does not prove
the range is reliable.

**Improve: use separate
data to choose models
and set ranges.**

Limited samples; no statistical significance test or independently verified coverage guarantee.

Next: test real water use at a new facility

Keep (Adopt) Tree models, physical limits, permit-location checks

Improve (Modify) Check neighbors, shared outages, and prediction ranges

Reject / Defer R2 is not confidence; wait on neural and generative LLM models

Will it work during drought at an unseen facility, with reliable prediction ranges?

Shaun: code + ranges Troy: data + drought Scott: permits + evidence